Shren Patel

Staff AI Engineer — Serverless Agentic Workflows on AWS

patel.shrenil@gmail.com · +1 613-551-9989 · Canada remote · NA-friendly · EST/PST
linkedin.com/in/shrenp · github.com/blopit · shrenil.com

SUMMARY

10+ years software across mature production environments — strong Python + Node.js. Built CentrOS — multi-agent orchestration with function/tool calling, RAG over pgvector + OpenSearch, streaming LLM responses, Langfuse evals, event-driven workflows; 3,980+ Vitest tests on main. Reliability and integration quality over novelty.

EXPERIENCE

Founder CentrOS — AI Agent Orchestration Platform *Feb 2026 – Present*
Remote · Canada • TypeScript · Hono · PostgreSQL · Drizzle · Vitest · LLM tool-calling

- Built CentrOS production AI workflow services in Python + Node.js — multi-agent orchestration, function/tool calling, RAG, streaming LLM responses across event-driven pipelines; 3,980+ tests on main.
- Implemented hybrid vector retrieval + streaming-response patterns (pgvector, OpenSearch-style) with chunk hygiene, freshness rules, source ranking — cut token spend 40% while improving grounding signal.
- Added Langfuse-style observability + synthetic AI testing: traces, replay cases, regression suites, token-cost reporting — caught 22+ regressions before release across CentrOS verticals.
- Built MCP-style tool registry with scoped OAuth permissions, JSON-schema validation, retry policy, side-effect logging — turned ad-hoc API calls into 100% audited typed agent tool invocations.
- Dogfooded CentrOS on 23-unit motel vertical (guest comms, refunds, vendor invoices, compliance) — 3,980+ Vitest tests across 347 files passing on main; CI-backed live ops.

Creator, WEBTOON Original — AXED WEBTOON *Jun 2018 – Dec 2020*
Remote • Python · OpenCV · image processing

- Built Python/JS image-processing + consistency-check pipelines for 445K+ subscriber WEBTOON Original at 2x/week ship cadence — 10% fewer re-uploads, hundreds of operator hours saved.
- Designed queue → tooling → review → ship operator workflow at 2M+ monthly views on WEBTOON Original — precursor to today's HITL agent + human production system at CentrOS.

iOS Mobile & Web Developer Coursera *Jan 2017 – Apr 2017*
Mountain View, CA • React · Python · iOS SDK · Swift · NLP/search

- Built Python search tooling lifting course content discovery 60% (pre-LLM NLP era) — same retrieval problem CentrOS now solves with embeddings, hybrid RAG, and pgvector.

Senior Frontend Developer Faire *Jun 2021 – Aug 2022*
Remote · Canada • React · TypeScript · MobX · Jest · Storybook · A/B testing

- Drove site-wide perf gains via A/B testing + staged rollouts; same guardrail discipline now applied to agent versioning, shadow traffic, and canary releases at CentrOS — 5+ wins shipped.

SKILLS

AI Workflow Engineering:

- Multi-agent orchestration · tool/function calling · streaming LLM responses
- RAG · embeddings · vector retrieval (OpenSearch · pgvector · FAISS)
- Langfuse-style tracing · synthetic AI testing · token-cost optimization · evals
- Workflow orchestration · state machines · event-driven async patterns

Languages:

- Python · Node.js · TypeScript

Cloud (AWS-heavy):

- AWS Lambda · API Gateway · DynamoDB · S3 · SQS · EventBridge · Step Functions
- Docker · serverless architectures · CI/CD · production logging & monitoring

EDUCATION

University of Waterloo Bachelor of Software Engineering (BSE) — Computer Software Engineering *2012 – 2018*
Waterloo, ON